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1. Title of the invention:

A region-first, medication-aware meal-planning system that detects and resolves conflicting dietary rules for patients living with multiple chronic diseases.

Working name of the prototype: MediPlate.

2. Field /Area of invention:

Medical informatics and clinical decision support systems; computer-implemented personalised clinical nutrition; food–drug interaction screening; combinatorial optimisation (mixed-integer linear programming) applied to diet formulation; applied machine learning for chronic-disease risk stratification.

Likely classification: G16H 20/60 (ICT for dietary therapy), G16H 50/20 (ICT for computer-aided diagnosis/decision support), G16H 20/10 (ICT for medication regimens), G06Q 50/22 (healthcare services), G06N 20/00 (machine learning).

3. Prior Patents and Publications from literature (provide a table summarizing the prior art)

Each item below was retrieved and read. Columns (a), (b) and (c) record whether the reference discloses the three features that this invention combines: **(a)** pre-filtering the food set against the patient's *entire* medicine list, **(b)** detecting and resolving *conflicting* dietary rules between co-existing diseases and explaining the resolution to the patient, **(c)** selecting dish, portion, salt level and preparation by mixed-integer linear programming under a *home-region-first* cuisine objective.

Ref.	Year / Owner	What it discloses	(a)	(b)	(c)	Gap
US10373522B2	2019, IBM	Genetic-algorithm meal generation from seed ingredients and exclusion criteria; fitness on cost, shelf life, flavour pairing, waste, preparation time	No	No	No	No medication logic, no disease rules, no clinical constraint optimisation
US12354728B2	2025, Bia Studios LLC	Meal building from on-hand ingredients and dietary needs; randomised sub-meal generators, allergy filtering, calorie split	No	No	No	Randomised, not optimised; no drug or comorbidity reasoning
US8060383B2	2011, Badinelli	Barcode/label scan compared against a stored profile that includes prescriptions, allergies and chronic illness; real-time alert	Partial	No	No	Point-of-purchase screening of one product; no plan generation or conflict resolution
US10902943B2	2021, IBM	Predicts drug–food interactions from drug/food similarity feature vectors and logistic regression	Partial	No	No	Predicts isolated drug–food pairs; is not a meal planner
US11580879B2	2023, Medtronic MiniMed	Personalised nutrition from a food ontology with linear optimisation (Simplex) and constraint satisfaction	No	No	Partial	Closest on mathematics; no medication filter, no conflict ledger, no region or preparation variables
US12254972B2	2025, Seekingsimple Inc.	"Nutritional prescription" templates per chronic condition, including patients with several conditions; removes harmful and adds beneficial products	No	Partial	No	Additive templates; no priority-based resolution, no optimiser, no drug screening
US6872077B2	2005, Solos Health LLC	Recipe templates with ingredient contribution values and nutritional priority values to meet macronutrient allowances	No	No	No	Macronutrient personalisation only; no clinical or pharmacological layer
Sklan & Dariel, Br J Nutr, 10.1079/bjn19930102	1993	Foundational mixed-integer linear programming formulation of human diet planning with whole-unit foods and frequency constraints	No	No	Partial	Establishes the solver class; eating practices are generic, not a regional preference ordering
Marashi-Hosseini et	2023	Mamdani fuzzy dietary decision-support	No	Parti	No	Blends competing

Ref.	Year / Owner	What it discloses	(a)	(b)	(c)	Gap
al., Sci Rep, 10.1038/s41598-023-39371-4		system, 1,144 rules, for co-existing obesity, diabetes, hypertension, hyperlipidaemia and kidney disease		al		requirements inside fuzzy membership functions; no clinical-priority ranking and no patient-facing explanation
Paidipati et al., IJERPH, 10.3390/ijerph18157842	2021	Pre-emptive and non-pre-emptive goal programming over 62 Indian recipes for six calorie levels in diabetes	No	No	Partial	Single disease; Indian foods treated as one pool, not regionally stratified
Ma et al., Foods, 10.3390/foods13132144	2024	Graph convolutional network over a user–recipe–nutrient graph for health-aware recipe recommendation	No	No	No	Learned ranking, not constrained optimisation; no clinical or drug constraints

4. Summary and background of the invention (Address the gap / Novelty)

Background and problem

About one in four Indian adults over 30 lives with more than one chronic disease at the same time, and more than 40% of people with diabetes also have high blood pressure. Existing diet applications plan for **one** disease at a time. This produces three failures that patients meet in practice:

- **Conflicting advice is left to the patient.** Kidney disease says "eat less protein and less potassium"; diabetes says "eat more fibre and pulses"; heart failure says "eat more potassium-rich fruit". A patient with all three receives three plans that contradict each other and no rule for deciding which instruction wins.
- **Medicines are ignored.** Warfarin fixes a vitamin-K band, ACE inhibitors and potassium-sparing diuretics make potassium-rich food dangerous, statins clash with grapefruit, MAO inhibitors with fermented and aged foods. Existing planners either store the prescription without acting on it, or score one drug–food pair at a time, and never remove the offending dish from the plan before it is offered.
- **The food is unfamiliar.** Plans are built from a national or western food pool, so a patient in Tamil Nadu is told to eat dishes from another region and abandons the plan. In our own measurement, without a region rule only 10.6% of a South-Indian patient's regional dishes came from South India.

Novelty: what is new here

The invention is the **combination** of three mechanisms in one pipeline, in this order, where the output of each one constrains the next:

1. **Multi-drug, multi-disease pre-filtering.** The patient's complete medicine list is expanded into drug classes and each class contributes food rules. Those rules are merged with the rules of every condition the patient has, and with diet type, allergies and religious choices, into a single removal set that is applied **before** any plan is generated, so an unsafe dish is never presented and then retracted.
2. **Conflict detection and priority-based resolution with a plain-language ledger.** Every condition and medicine contributes numeric targets carrying a clinical priority (1 = immediate safety, 2 = organ protection, 3 = disease control, 4 = weight and general, 5 = wellness). When two targets on the same nutrient disagree, the stricter target from the higher priority is kept, the discarded target is recorded, and the system emits a patient-readable sentence such as "Sodium (salt): the strictest limit (2,000 mg from high blood pressure) was used." The ledger is part of the output, not an internal log.
3. **Region-first optimisation over dish, portion, salt level and preparation.** The surviving dishes are given to a mixed-integer linear program that fills nine meal slots a day. The patient's state is mapped to a home food zone; the program searches that zone first, widens to neighbouring zones and then to all-India dishes only when the limits cannot otherwise be met, and treats **salt level** (full / half / quarter) and **potassium-leaching preparation** as decision variables alongside dish and portion.

No reference found in the search (Section 3) discloses this chain. US11580879B2 optimises but ignores medicines and conflicts; Marashi-Hosseini (2023) addresses comorbidity but dissolves conflicts inside fuzzy logic without explanation; US10902943B2 predicts drug–food pairs but does not plan meals; Paidipati (2021) optimises Indian menus for a single disease with no regional stratification. The unclaimed combination is a **full-regimen drug filter**

feeding a priority-based conflict resolver that emits an explanation, whose surviving food set is optimised region-first with preparation-level decision variables.

5. Objective(s) of Invention

4. To generate a weekly meal plan that satisfies the dietary limits of **all** of a patient's chronic diseases simultaneously, rather than one disease at a time.
5. To remove, before planning, every dish that interacts with any medicine in the patient's current regimen.
6. To detect conflicting dietary requirements between co-existing diseases, resolve them by clinical priority, and disclose each resolution to the patient in ordinary language.
7. To keep the plan culturally acceptable by drawing food from the patient's own region of India before any other region.
8. To use preparation-level controls — salt reduction and potassium leaching — as a way of keeping a familiar dish in the plan instead of deleting it.
9. To explain every dish, every limit and every removal, so that a clinician can audit the plan and a patient can trust it.
10. To adapt to the individual over time from meal logs, ratings and after-meal blood-sugar readings.
11. To deliver all of the above in the patient's own language (English, Tamil and Hindi in the prototype) on an ordinary web browser.

6. Working principle of the invent (in brief)

The system converts a patient record into a constrained optimisation problem and solves it, then converts the solution back into explanations. Five stages run on every request:

Stage	Name	What it does
E1	Profile builder	Derives BMI on Indian cut-offs, eGFR by CKD-EPI 2021, kidney stage, reference body weight and energy requirement (Mifflin-St Jeor). Turns on kidney-safe limits automatically when eGFR < 60.
E2	Safety filter	Removes dishes failing diet type, allergies, condition rules, medicine rules, or a hard limit even at the smallest salt-reduced portion.
E3	Target builder and conflict resolver	Collects numeric targets from every condition, medicine and risk result with a clinical priority; merges them; resolves clashes; writes the conflict ledger.
E4	Optimiser	Mixed-integer linear program choosing dish × portion × salt level × leaching for nine slots a day, home region first, hard limits as constraints and soft goals as penalties.
E5	Explainer	Produces the reason for each dish, region of origin, day totals against every limit, meal sugar load, removed dishes and medicine timing, in the patient's language.

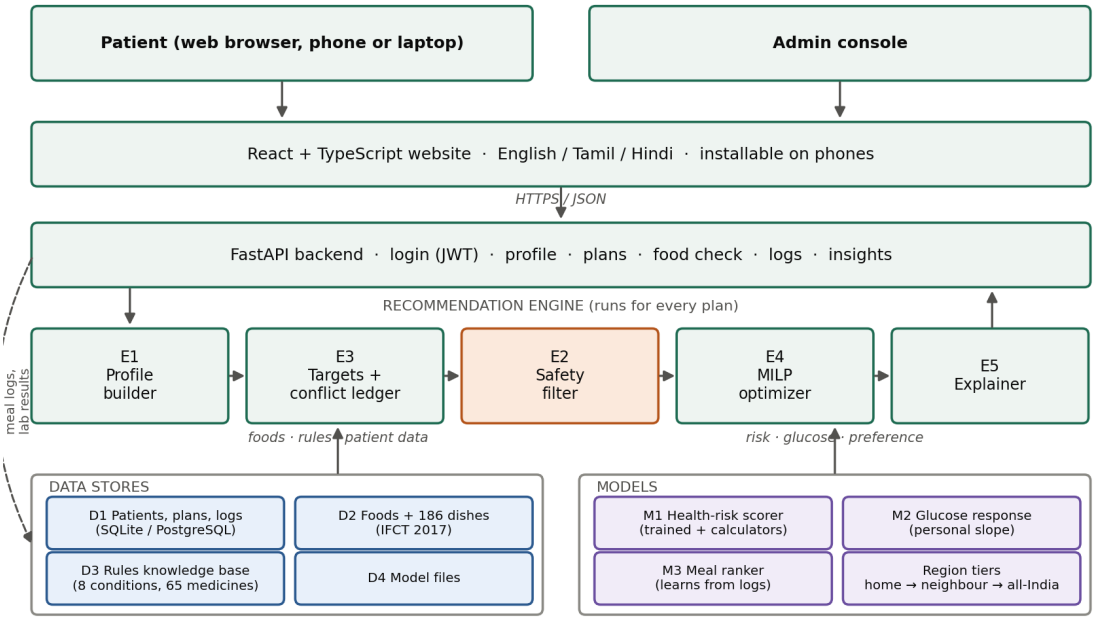


Figure 1. System architecture. Every plan request passes through the five stages; logs return to the learning models.

7. Description of the invention in detail (Include drawing and or photograph as needed)

7.1 Knowledge base and data preparation

An offline pipeline builds the data the system runs on and refuses to publish a broken build. It downloads the Indian Food Composition Tables 2017 (542 foods) and gap-fills 11 foods from USDA SR Legacy, giving **553 foods**; converts all units to grams per 100 g and kilocalories; runs an Atwater consistency check that caught one published entry whose stated energy was double its macronutrients; costs **186 regional dishes** across five food zones into nutrients per serving; compiles and schema-checks the rule base — **8 diseases, 65 medicines in 35 drug classes, 23 food tags**; and trains the risk model. All explanations are stored as **195 message keys in English, Tamil and Hindi**, so the engine emits keys and the interface renders language.

The 36 states and union territories are mapped to five food zones (North, South, East, West, North-East) with an explicit list of neighbouring zones per state, which is what makes the widening search in E4 possible.

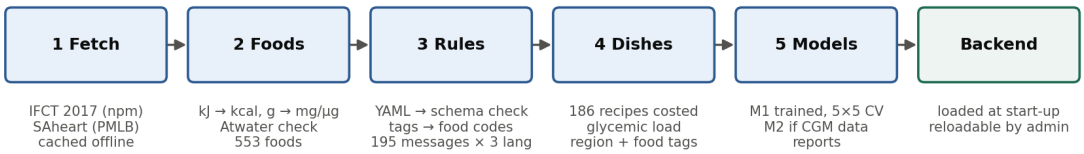


Figure 2. Offline data pipeline: fetch, clean, merge, validate, train.

7.2 Stage E1 — patient context

From age, sex, height, weight, waist, activity, state, diet type, conditions, medicines and laboratory results the system derives BMI against Indian cut-offs (overweight ≥ 23, obese ≥ 25), eGFR by the CKD-EPI 2021 race-free equation, kidney stage, and a reference body weight capped at BMI 25 that is used for every per-kilogram target so that protein limits are not inflated by excess weight. Energy is Mifflin-St Jeor times an activity factor, minus 500 kcal where weight loss is indicated. A kidney stage derived from laboratory results switches on kidney limits even if the patient did not report kidney disease.

7.3 Stage E2 — pre-filter against the whole regimen

Each medicine is resolved to a drug class, and each class contributes food rules. The rules of every condition, the diet type, the allergy list and the beef/pork choice are unioned into one removal set, and each removed dish carries the reason for its removal. A dish is also removed if it breaks a hard limit even at its smallest portion with quarter salt, so an impossible dish never enters the optimiser. Examples from the implemented rule base:

Medicine or condition	Rule applied	Effect on the food set
Warfarin	Vitamin K held in a narrow daily band rather than avoided	Very-high vitamin-K dishes removed; the rest bounded above and below
ACE inhibitor, ARB or potassium-sparing diuretic	Potassium ceiling; salt substitutes forbidden	High-potassium dishes limited; a no-salt-substitute advice line is emitted
Statin (simvastatin, atorvastatin)	Grapefruit and related citrus avoided	Interacting dishes removed
MAO inhibitor	Tyramine-bearing aged, fermented and cured foods avoided	Interacting dishes removed
Metformin	Alcohol avoided; medicine taken with meals	Timing instruction attached to the plan
Chronic kidney disease stage ≥ 4 or serum $K^+ > 5.0$	Potassium ≤ 2,000 mg/day, phosphorus ≤ 800 mg/day, both hard	Star fruit and tender coconut water removed outright; leaching enabled
Gout	High-purine foods avoided; red meat heavily penalised; alcohol avoided	Organ meat and high-purine dishes removed

7.4 Stage E3 — targets, conflict detection and the ledger

A target is a tuple {nutrient, bound, value, hardness, priority, reason}, where the value may be absolute, per kilogram of reference weight, a percentage of energy, or per 1,000 kcal, and may be conditional on a laboratory result or disease stage. Targets are gathered from every condition, every medicine, the risk model and the healthy-eating default. Two targets on the same nutrient and bound conflict; the resolver keeps the one with the higher clinical priority, breaks ties by strictness, and records the discarded target with the reason for its discard. This record is rendered as the conflict ledger. A worked example produced by the prototype for a patient with chronic kidney disease stage 4 and high blood pressure:

> Conflict ledger: "Potassium: high blood pressure advice suggests at least 3,500 mg a day, but kidney disease at stage 4 needs it below 2,000 mg. The kidney limit was used, because it is the immediate safety limit." · "Sodium (salt): the strictest limit (2,000 mg from high blood pressure) was used."

The priority scale is: **1** immediate safety (for example a potassium ceiling with a raised serum potassium), **2** organ protection (protein restriction in kidney disease), **3** disease control (carbohydrate and sugar limits in diabetes), **4** weight and general health, **5** wellness. Because the priority is attached to the rule rather than hard-coded in the engine, a clinician can revise the knowledge base without touching the software.

7.5 Stage E4 — the region-first mixed-integer program

Let a variant be a triple of dish d, portion p and salt level s ∈ {1, 0.5, 0.25}, with a further binary for potassium leaching where the preparation permits it (leaching is disallowed on raw dishes such as salads, curd and sprouts). Let x be 1 when variant v is placed in slot j of day t. For nine slots a day:

$$\text{minimise} \;\; \Sigma\, \text{penalty}(v)\cdot x[v,j,t] + \Sigma\, w(g)\cdot \text{shortfall}(g,t) \quad \text{subject to}$$

$$\Sigma\, x[v,j,t] = 1 \text{ for every slot } j; \quad \Sigma\, \text{nutrient}(n,v)\cdot x[v,j,t] \le \text{limit}(n) \;\; \text{for every hard limit } n;$$

$$\Sigma\, x \text{ over the week} \le 5 \text{ for any one dish;} \quad \text{carbohydrate(meal)} \ge \text{floor;} \quad \text{vitamin K within band}$$

Hard limits become constraints, so they cannot be broken; soft goals become penalised shortfalls, so the solver trades them against each other instead of failing. The penalty term carries the region preference: dishes from the patient's home zone carry no penalty, neighbouring zones carry a penalty, all-India dishes carry a larger one, and the program only reaches them when the constraints cannot otherwise be satisfied. Repetition is capped so that the week stays varied. The prototype solves with the CBC mixed-integer solver through OR-Tools at a 5% relative gap.

Salt level and leaching are inside the optimisation rather than applied afterwards, which is what lets a familiar dish survive a strict limit: a South-Indian sambar that fails a 2,000 mg sodium ceiling at full salt is kept at half salt, and a

potato dish that fails a potassium ceiling is kept with a leaching instruction (soak and discard the water, retention factor 0.6) instead of being deleted.

7.6 Stage E5 — explanation

For every dish the output carries the reason it was chosen, its region of origin, the portion, the salt level and any preparation note; for every day it carries each nutrient total against its limit with the source of that limit; and for the patient it carries the conflict ledger, the list of removed dishes with reasons, medicine timing instructions and the per-meal sugar load. The server sends message keys and parameters, so the same plan renders in English, Tamil or Hindi.

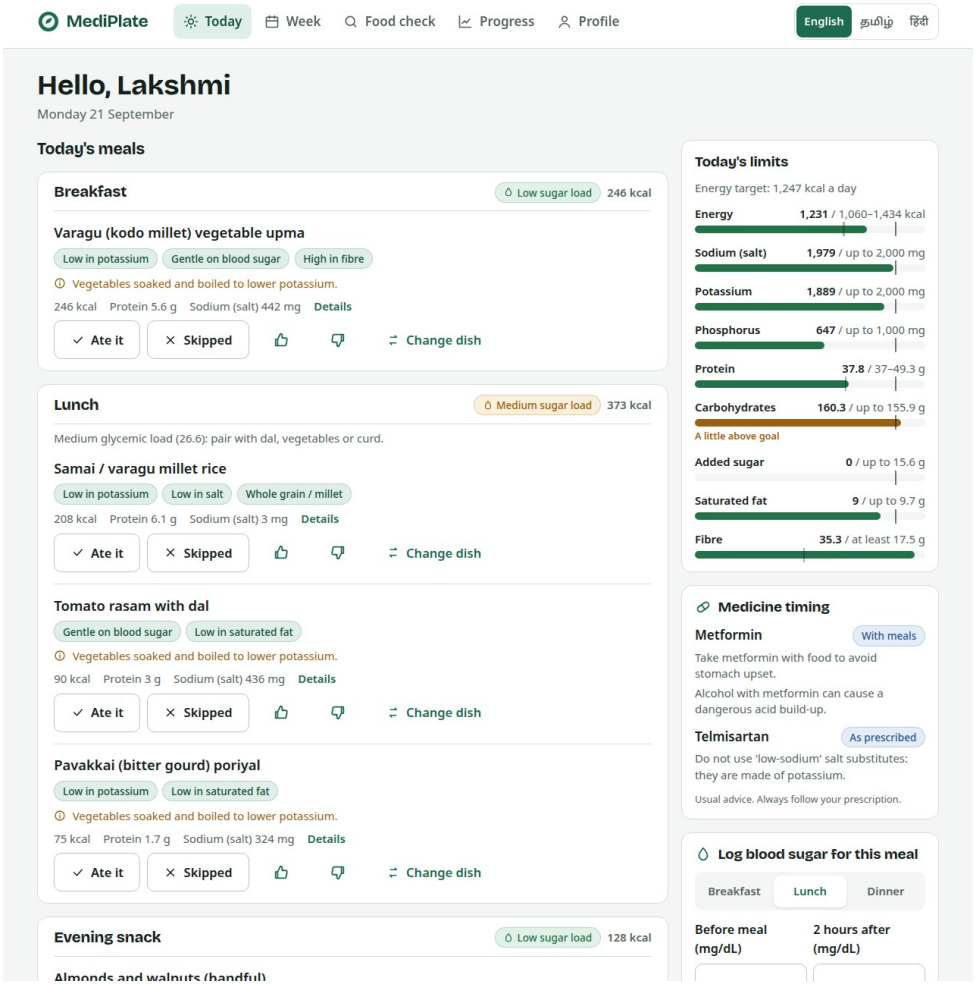


Figure 3. Prototype interface: the day's meals with reasons, limit meters, medicine timing and logging controls.

7.7 Learning components

M1, a logistic-regression heart-risk indicator trained on 462 adults, together with validated calculators (FIB-4, Indian Diabetes Risk Score, IDF metabolic-syndrome criteria), raises the priority of cardiovascular limits when risk is high. **M2** estimates the patient's own blood-sugar response: glycemic load is used until the patient has logged three before/after-meal pairs, after which a personal slope of sugar rise per 10 g of carbohydrate is fitted and used in planning. **M3** learns from eaten, skipped, liked and disliked meals at the level of ingredients, so disliking one dish lowers every dish sharing its main ingredient. None of the three may override a hard limit; they act only on the penalty terms.

8. Experimental validation results:

A working full-stack prototype was built and measured. Every number below is produced by a reproducible evaluation script over the implemented system, not by simulation of a design.

8.1 Safety of generated plans

The planner was run for **100 synthetic patients** drawn from a fixed seed, covering all 36 states, three diet types, one to four co-existing conditions each, kidney stages 2–5, random potassium and phosphorus results and 14 medicine

combinations, for three days each (**300 planned days**). Each day was then re-totalled independently from the raw dish data rather than from the optimiser's own numbers.

Measure	Result
Days for which a feasible plan was produced	300 of 300
Hard-limit checks performed (daily limits, meal carbohydrate floors, vitamin-K band, energy window, removed dishes)	1032
Violations	0
Days satisfied inside the home region (tier 1)	291
Days needing a neighbouring region (tier 2)	9
Mean share of regional dishes from the home zone	99.2%
Patients whose limits required conflict resolution	12 of 100
Optimiser time per day (median / 90th percentile / maximum, 2 cores)	0.37 s / 1.76 s / 2.56 s

8.2 Effect of the region-first mechanism (ablation)

The same patients were planned twice, once with the region preference and once with it removed, and the share of regional dishes coming from the patient's own zone was measured.

Home zone	With region-first	Without region-first
North	100.0%	47.1%
South	100.0%	10.6%
East	100.0%	17.1%
West	100.0%	29.5%
North-East	100.0%	9.3%

The mechanism is therefore responsible for the whole of the cultural fit; without it the North-East patient receives 9.3% of regional dishes from the North-East.

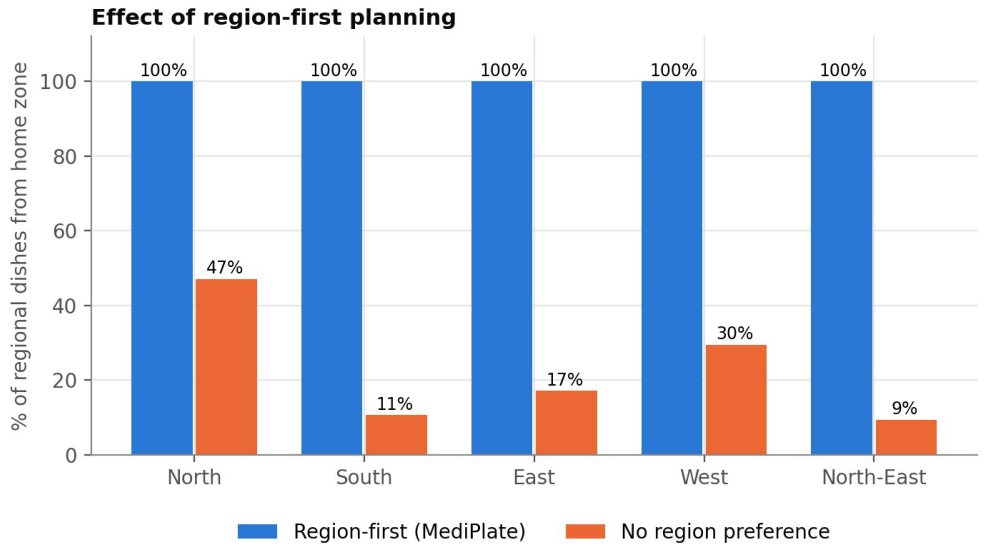


Figure 4. Share of regional dishes from the patient's home zone, with and without region-first planning.

8.3 Pre-filter behaviour

Dishes removed before planning, by scenario, out of 186:

Patient scenario	Dishes removed	Dishes limited
No conditions	0	0
Vegetarian	25	0
Peanut + milk allergy	54	0

Patient scenario	Dishes removed	Dishes limited
Gout	1	0
Kidney disease stage 4	1	0
Warfarin	8	0
Simvastatin	1	0
Diabetes + BP + CKD 4 + warfarin	9	5

8.4 Risk model (M1)

Trained on the South African Heart Disease dataset (**462 adults**, 160 with coronary heart disease, 6 features), logistic regression was selected over gradient boosting, which overfitted. All figures at the 0.5 decision threshold.

Metric	Training data	5×5-fold cross-validation	Held-out test (30%)
Accuracy	0.723	0.713 ± 0.040	0.647
Precision	0.621	0.608 ± 0.079	0.489
Recall	0.512	0.499 ± 0.074	0.458
F1	0.562	0.545 ± 0.066	0.473
ROC AUC	0.779	0.769 ± 0.039	0.756

The model acts only as a caution flag that tightens cardiovascular limits; it is not used for diagnosis and cannot override a hard limit.

8.5 Software verification

91 automated backend tests, including a property test that plans for 36 randomly generated patients across all states, diets, condition combinations, kidney stages, laboratory results and 14 medicine groups and then re-derives every day from raw data; a strict type-checked web build that fails on any missing translation; and 5 browser tests covering sign-up, five-step onboarding, plan generation, meal logging, dish swap, food check in Tamil script, language switching, mobile layout and the administration console. All pass.

9. What aspect(s) of the invention need(s) protection?

Protection is sought for the following aspects, in order of importance.

A. Principal aspect (system and method)

A computer-implemented method of generating a dietary plan for a patient having **two or more** co-existing chronic conditions and a medication regimen of **one or more** medicines, comprising: resolving each medicine to a drug class and deriving food rules from it; forming a removal set from the union of the drug-class rules, the condition rules and the patient's dietary constraints, and excluding the corresponding foods from a candidate set **before** plan generation; deriving numeric nutrient targets each carrying a clinical priority; **detecting a conflict** between two targets on the same nutrient, selecting one by clinical priority, and **emitting a human-readable record of the resolution as part of the output**; and selecting, by mixed-integer optimisation over the reduced candidate set, a combination of dish, portion, salt level and preparation method for each meal slot subject to the resolved targets as hard constraints.

B. Region-first selection

The above, wherein candidate dishes are stratified into geographic food zones, the patient's residence is mapped to a home zone, and the optimisation applies a penalty ordering that exhausts the home zone before neighbouring zones and neighbouring zones before the remaining set, widening only when the hard constraints cannot otherwise be satisfied.

C. Preparation-level decision variables

The above, wherein salt level and a potassium-reducing preparation (leaching with a retention factor) are decision variables within the same optimisation, and a preparation variable is permitted only for dishes marked as cooked, so that a dish that would fail a limit at full salt or without leaching is retained in modified form rather than removed.

D. Further aspects

- Automatic activation of organ-protective limits from laboratory results (eGFR, serum potassium, serum phosphorus) even where the patient has not reported the condition.
- Use of a reference body weight capped at a target BMI for all per-kilogram targets.
- Removal of a candidate dish when it breaches a hard limit at its smallest portion and lowest salt level, evaluated before optimisation.
- Bounding rather than excluding a nutrient where a medicine requires stability (vitamin K under warfarin).
- Ingredient-level propagation of a patient's dislike to other dishes sharing that ingredient.
- Progressive personalisation of the glycemic response from a small number of paired readings, bounded so that it cannot relax a hard limit.
- Emission of language-independent message keys with translated parameters, so that the clinical explanation is reproduced exactly in each supported language.

Disclosure status (to be confirmed by the inventor before submission): no paper has been published, presented or submitted, and the work has not been offered for sale. The prototype source is held in a GitHub repository — if that repository is public, the date of its first public commit is a public disclosure and must be stated here.

10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

Research			Development			Deployment		
TRL 1	**TRL 2**	**TRL 3**	**TRL 4**	**TRL 5**	**TRL 6**	**TRL 7**	**TRL 8**	**TRL 9**
Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant environment	Technology demonstrated in a relevant environment	System prototype demonstration in an operational environment	System complete and qualified	Actual system proven in an operational environment
			✓					

TRL 4 — technology validated in a laboratory. A complete, working prototype of the whole pipeline exists and has been validated on 300 planned days for 100 synthetic patients with 0 violations in 1,032 hard-limit checks, on a fixed and reproducible test bench. It has not yet been used by real patients or evaluated by a registered dietitian and a pharmacist, which is what TRL 5 would require. The clinical knowledge base is a careful prototype compiled from ICMR-NIN 2024, ADA 2025, KDIGO 2024, AHA/DASH, EASL 2024, ACR 2020 and drug labels, and is marked throughout as requiring clinical review before patient use.

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